

ZoneCast

Deployment guide for a blank machine
Ubuntu Server 24.04 LTS / 26.04 LTS

Technical manual — installation, architecture, configuration, migration and security
Application version 1.5.4 — September 2026

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1. Introduction and what the package includes

ZoneCast is a PA/intercom management system for IP speakers (Fanvil A233 and, via the generic driver, other multicast brands such as Algo, CyberData, Grandstream, Axis, Tiptel, Akuvox): zones, schedules with holidays from multiple countries, multicast RTP playback, announcement volume for supported brands, device configuration backups, persistent logs, two-factor authentication, host resource monitoring, light/dark/automatic theme, multilingual interface, and full data export/import.

This guide covers installation on a **brand-new Ubuntu Server LTS machine** ("blank"), both with Docker and natively (systemd), as well as migrating data from an existing installation.

WHICH METHOD SHOULD I CHOOSE?

If the machine is **dedicated exclusively** to ZoneCast, the **native install (Method A)** is recommended: multicast RTP traffic reaches the physical network without having to work around the Docker bridge's NAT, and system time/network management works without the AppArmor limitations tied to containers. If the machine also hosts other services, or you'd rather be able to roll back quickly to a previous image, use Docker (Method B).

2. Architecture and components used

ZoneCast is a monolithic Python web application, designed to run on a single dedicated machine. The diagram below summarizes the main layers:

```
Browser (Bootstrap 5.3 + vanilla JS)
| HTTP/JSON
v
FastAPI (app/main.py) ---- routers (auth, speakers, zones, media,
|                               schedules, system, backups, logs)
v
services/ (application logic: scheduler, device drivers,
|         audio_analysis, network_config, system_time, alerts...)
v
SQLAlchemy ORM ----> SQLite (data/zonecast.db)

APScheduler (in-process) runs the scheduled playbacks
and invokes the device drivers to send multicast/SIP
```

Main software components

Component	Role
FastAPI + Uvicorn	Asynchronous web framework and ASGI server
SQLAlchemy + SQLite	Application ORM and database (single file, no external DB server)
Jinja2	Server-side page rendering (dashboard.html, login.html)
Bootstrap 5.3 (CDN)	UI components and native light/dark theme support (data-bs-theme)
Vanilla JS (app.js, i18n.js, theme.js)	Frontend logic, translations (7 languages), light/dark/automatic theme
APScheduler	In-process scheduling engine for planned playbacks

Component	Role
holidays (Python library)	Per-country holiday calendars (IT/US/GB/FR/DE/ES/JP/CN), per schedule
httpx	Async HTTP client for Fanvil device provisioning
cryptography, bcrypt, pyotp	Sensitive-data encryption, password hashing, TOTP for 2FA
netplan (host)	Machine network configuration, managed by ZoneCast on request
chrony or systemd-timesyncd (host)	Time sync — ZoneCast auto-detects which of the two is installed (chrony on 26.04+, timesyncd on 24.04)

The **driver** pattern (app/services/drivers/) abstracts away the differences between speaker brands: each driver declares which features it supports (automatic multicast push, config backup, announcement volume) via simple boolean flags, so the interface automatically adapts to the selected device with no conditional code scattered across the application.

3. Machine requirements

Requirement	Recommended value
Operating system	Ubuntu Server 24.04 LTS or 26.04 LTS, minimal install
CPU	2 vCPU (1 is enough for small installations)
RAM	2 GB minimum, 4 GB recommended
Disk	20 GB minimum (more if you plan to store a lot of audio files)
Network	Static IP on the same LAN as the speakers, with multicast routing enabled (no router/switch filtering IGMP)
Access	A user with sudo privileges reachable via SSH

4. Preparing the blank machine

After installing Ubuntu Server (from the official installer, with OpenSSH enabled) and logging in over SSH for the first time:

```
# Update system packages (important: use dist-upgrade, not just
# upgrade, especially before any future major Ubuntu version upgrade)
sudo apt update && sudo apt dist-upgrade -y
sudo reboot    # if the kernel was updated

# Check the static IP address (usually configured during install
# via netplan, e.g. /etc/netplan/00-installer-config.yaml)
ip addr show

# Check that the date/time is correct and in sync
timedatectl
```

WARNING

If the machine doesn't have a static IP, speakers configured with the server's address will stop working the first time it reboots with a different IP. Configure the static IP in `/etc/netplan/*.yaml` before proceeding — or leave it to the installer and manage the final IP later from ZoneCast's own System panel (section 9).

5. Method A — Native installation (systemd, recommended)

Copy the package onto the machine and run the provided install script (deploy/install_ubuntu.sh):

```
scp zonecast_install_kit.tar.gz user@new-machine:/tmp/
ssh user@new-machine
mkdir -p /tmp/zonecast-src
tar -xzf /tmp/zonecast_install_kit.tar.gz -C /tmp/zonecast-src
cd /tmp/zonecast-src
sudo ./deploy/install_ubuntu.sh
```

In order, the script:

- 1 Installs the required system packages (Python 3, ffmpeg, rsync, git, sqlite3, curl)
- 2 Creates a dedicated, unprivileged system user, zonecast
- 3 Copies the application to /opt/zonecast
- 4 Generates a .env file with a random SECRET_KEY, if one doesn't already exist
- 5 Creates a Python virtualenv (against the current system interpreter — important after a major Ubuntu upgrade, see note below) and installs the dependencies
- 6 Installs and starts the systemd service (deploy/zonecast.service)
- 7 Installs the privileged script deploy/zonecast-netctl.sh with a dedicated sudoers rule (see section 11)

NOTE — OS upgrades

If you later upgrade Ubuntu's major version (e.g. 24.04 → 26.04), the default system python changes version: the existing venv will stop working ("ModuleNotFoundError") because its symlink still points at the previous interpreter. Recreate the venv against the current python: `sudo -u zonecast python3 -m venv --clear /opt/zonecast/venv && sudo -u zonecast /opt/zonecast/venv/bin/pip install -r /opt/zonecast/requirements.txt`, then restart the service.

Once done, check its status:

```
sudo systemctl status zonecast
sudo journalctl -u zonecast -f # live log
```

6. Method B — Docker installation

```
sudo apt update
sudo apt install -y ca-certificates curl gnupg
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER # then log back in over SSH

tar -xzf zonecast_install_kit.tar.gz -C ~/zonecast
cd ~/zonecast
cp .env.example .env
nano .env # customize SECRET_KEY, admin credentials, etc.
docker compose up -d --build
```

docker-compose.yml uses `network_mode: host`, essential so multicast RTP traffic actually reaches the physical LAN. Works only on a Linux host. In this mode the System panel shows Network/NTP as read-only (netplan and systemd-timesyncd/chrony manage the host, not the container).

7. Application configuration (.env)

Variable	Description
SECRET_KEY	Session-signing key — generated automatically by the native install script
DEFAULT_ADMIN_USERNAME / DEFAULT_ADMIN_PASSWORD	First admin's credentials, created only if the database is empty. Change the password on first login
DATABASE_URL	Path to the SQLite database (normally left as-is)
GLOBAL_ALL_CALL_ADDRESS / _PORT	Multicast address/port for the "all speakers" announcement
RTP_PAYLOAD_TYPE	G.711 encoding: 0 = PCMU, 8 = PCMA — depends on the devices
TIMEZONE	Timezone used by the scheduler (default Europe/Rome)
APP_PORT	Listening port (80/443 require CAP_NET_BIND_SERVICE, already granted in the native service)
MAX_UPLOAD_MB / MAX_DURATION_SECONDS	Limits on uploadable audio files

After editing .env: `sudo systemctl restart zonecast` (native) or `docker compose up -d --force-recreate` (Docker).

8. First login and securing the account

- 1 Open `http://<machine-ip>` in a browser (default port 80, or whatever APP_PORT is set to)
- 2 Log in with the credentials set in DEFAULT_ADMIN_USERNAME / DEFAULT_ADMIN_PASSWORD
- 3 Change the password immediately from the Password button at the top (minimum 8 characters)
- 4 Consider enabling 2FA from the Security button
- 5 Create operational users from Users (admins only), with the appropriate role
- 6 Choose language and theme from the top bar — these are a per-browser personal preference, available to any user

9. Feature overview

Light / dark / automatic theme

Selectable by anyone from the top bar. Automatic computes real sunrise/sunset via the browser's geolocation (falling back to Rome if not granted), not just the OS preference.

Multilingual interface

English (default), Italian, French, German, Spanish, Japanese, Simplified Chinese. Preference saved in the browser, not shared between users.

Per-country holidays, per schedule

Each schedule can use the holiday calendar of Italy, the United States, the United Kingdom, France, Germany, Spain, Japan, or China, independently of the interface language.

Host resource monitoring

Disk, RAM and CPU in real time in the System panel, with a color-coded bar (green/yellow/red) based on usage.

First-login alerts

Automatic banner if there were failed schedules in the last 24 hours or disk space is running low.

Speaker brand dropdown

A list of the main multicast brands with an "Other" option for unlisted devices, instead of a free-text field.

Audio analysis

Every uploaded file is analyzed for bass/mid/treble balance; a red badge warns if the bass content is high and might sound poor on a PA horn.

Version and changelog

Clickable version number in System > Information, opens the change history.

10. Migrating data from another installation

Step 1 — Export from the source installation

From the existing dashboard: System → Export full configuration, set a strong password and download the generated .zcbundle file. It contains the database, the encryption key, media and device backups — treat it as sensitive material.

Step 2 — Import on the new machine, BEFORE the service's first start

```
# --- Native install ---
sudo systemctl stop zonecast
cd /opt/zonecast
sudo -u zonecast venv/bin/python -m app.tools.import_bundle /path/to/export.zcbundle
sudo systemctl start zonecast

# --- Docker install ---
cd ~/zonecast && docker compose stop zonecast
docker run --rm -v $(pwd)/data:/app/data -v $(pwd)/media:/app/media \
-v $(pwd)/backups:/app/backups -v /path/to/export.zcbundle:/tmp/export.zcbundle \
zonecast-zonecast:latest python -m app.tools.import_bundle /tmp/export.zcbundle
--yes
docker compose up -d
```

IMPORTANT

The import must be done with the service **stopped**: overwriting the SQLite database while the app has open connections risks corrupting it.

11. Security: 2FA, firewall, HTTPS, systemd hardening

Two-factor authentication (2FA)

Any user can enable it from the Security button: QR code for an authenticator app, 6-digit code, 10 recovery codes shown only once.

Firewall (ufw)

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp # or whatever port APP_PORT is set to
sudo ufw enable
```

How privilege escalation works for Network/Time

The application process always runs as the unprivileged zonecast user. Actions that need root privileges (changing the IP, timezone, NTP servers) go through a single script `deploy/zonecast-netctl.sh`, invoked via `sudo -n` and authorized by a sudoers rule **scoped to that one script** (never arbitrary commands).

TECHNICAL NOTE — systemd hardening and sudo

The `deploy/zonecast.service` file deliberately does **not** set `NoNewPrivileges` or `ProtectSystem=strict`, contrary to the more common hardening advice: both would prevent `sudo` from escalating privileges inside it (the former blocks `setuid` escalation outright; the latter would make `/etc` read-only even for the root child process `sudo` creates, since mount namespaces are inherited by child processes). The real security boundary is the `sudoers` rule, scoped to a single auditable script — not these settings, which here would directly conflict with the intended behavior.

HTTPS (optional)

The app speaks plain HTTP. To add TLS, put `nginx` in front of the app using the provided `deploy/nginx-zonecast.conf` template, then `certbot` for a Let's Encrypt certificate (with a domain) or a self-signed certificate (LAN IP only).

12. Day-to-day operations and useful commands

Action	Native (systemd)	Docker
Status	systemctl status zonecast	docker compose ps
Live log	journalctl -u zonecast -f	docker compose logs -f zonecast
Restart	systemctl restart zonecast	docker compose restart zonecast
Stop	systemctl stop zonecast	docker compose down

Application logs (logins, provisioning errors, playbacks) are always available and exportable from the dashboard: Logs section (admins only).

13. Backup and maintenance

- **Individual device config backup:** from each speaker's card, Backup button
- **Full application data backup:** System → Export full configuration — schedule a periodic export and keep it off the machine
- **Database-only backup:** `cp /opt/zonecast/data/zonecast.db /opt/zonecast/data/zonecast.db.bak-$(date +%Y%m%d)`

14. Common troubleshooting

Symptom	Possible cause / fix
Audio doesn't reach the speakers (Docker)	Check network_mode: host in docker-compose.yml and that the switch isn't filtering multicast/IGMP traffic
"The device did not return a login nonce"	The Fanvil's web server is temporarily overloaded or rate-limiting — wait a few minutes and retry from Apply now. Don't send rapid repeated requests to the same device
A changed IP stays reachable at the old address too	Check that no other file in /etc/netplan/ declares the same address besides 90-zonecast.yaml — ZoneCast disables pre-existing files on first apply, but an installation with multiple network interfaces may need manual review
NTP server set but status stays "Not synced"	Normal for the first 30-90 seconds after changing servers: chrony needs a few polling cycles before declaring itself synced. Also check the hostname is correct (e.g. time.google.com, not ntp.google.com — the latter doesn't exist)
Changing the NTP server fails with "systemd-timesyncd.service not found"	Fixed in version 1.5.0: NTP backend detection (chrony vs timesyncd) is now based on the service being installed, not on whether it's currently active
An operator user gets "Cannot read properties of null" saving a schedule	Fixed in version 1.5.1: a UI update was trying to remove admin-only nav items a second time
An operator user can't create/edit zones or speakers	Changed in version 1.5.2: operators and admins now have equal permissions on zones/speakers (including moving a speaker between zones) — only direct device actions (Apply now, Backup) and the Users/System/Logs/Backup sections stay admin-only
Error importing a bundle	"Wrong password or corrupted file": check you used the same password chosen at export time; the service must be stopped during import
Locked out after losing the 2FA authenticator app	Use one of the 10 recovery codes saved at activation; if lost, an administrator with server access can disable a user's 2FA from the database

15. Appendix — list of files provided

File	Purpose
zonecast_install_kit.tar.gz	Full application package — extract it on the new machine
deploy/install_ubuntu.sh	Native install script (Method A)
deploy/zonecast.service	systemd unit file for the native run
deploy/zonecast-netctl.sh	Privileged script for NTP/timezone/network, invoked via a scoped sudo rule
deploy/nginx-zonecast.conf	nginx reverse-proxy template for adding HTTPS
Dockerfile / docker-compose.yml	Image and orchestration for Method B
.env.example	Configuration file template
app/tools/import_bundle.py	Script to restore a full export
README.md	Project's technical reference documentation